# Adverse Drug Reactions (ADRs)

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# Safety of medicines – adverse drug reactions (ADR)

**Key facts**

- *Adverse effect is defined as any undesirable or unwanted, noxious and harmful effect of a drug. ADR occurs at doses normally used in humans for prophylaxis, diagnosis or therapy of disease, or for modification of physiological*
- *ADRs are among the leading causes of death in many countries.*
- *The majority of adverse drug reactions (ADR) are preventable.*
- *People in every country are affected by ADRs.*
- *In some countries, ADR-related costs such as hospitalization, surgery and lost productivity exceed the cost of medications.*
- *No medicine is risk-free. Vigilant assessment of the risks and benefits of medicines promotes patient safety.*

# ADVERSE EFFECT

*Adverse effect is defined as any undesirable or unwanted effect of a drug. The WHO suggested definition of adverse drug reactions (ADR) and adverse event (AE) as follows:*

**ADRs***: Any response which is noxious, unintended and which occurs at doses normally used in humans for prophylaxis, diagnosis or therapy of disease, or for modification of physiological function.*

**AE:** *Any untoward medical occurrence that may present during treatment with a pharmaceutical product but which does not necessarily have causal relationship with the treatment*

**Safety of medicines is an essential part of patient safety. Global drug safety depends on strong national systems that:**

*Monitor the development and quality of medicines.*

*Report their harmful effects.*

*Provide accurate information for their safe use.*

*Preventing and detecting adverse effects from medicines is termed* **pharmacovigilance***. Vigilant assessment of the risks and benefits of medicines applies throughout the life cycle of a medicine, from the pre-approval stage to use by patients.*
*Global information-sharing on adverse effects strengthens drug safety in countries, and can translate into timely policy decisions that safeguard patient safety when problems emerge.*

**Many ADRs (in some instances as many as 60%) are preventable, and may be due to a variety of reasons such as:**

- *Incorrect diagnosis of the patient's medical condition;*
- *Prescription of the inappropriate drug or incorrect dosage of the appropriate drug;*
- *An undetected medical, genetic or allergic condition that may cause a patient reaction;*
- *Self-medication with prescription medicines; not following the instructions for taking the medication;*
- *Interactions with other drugs (including traditional medicines) and certain foods;*
- *Use of sub-standard medication with composition and ingredients which do not meet scientific requirements, causing them to be ineffective and often hazardous;*
- *Use of counterfeit medicines with no active ingredients or inappropriate ingredients, which can be dangerous or fatal.*

# Risk Factors for Adverse Drug Reactions

*Understanding the risk factors for adverse drug reactions (ADRs) helps you manage medication safety effectively. Specific patient and drug-related factors contribute significantly to the likelihood of experiencing ADRs.*

## Patient-Related Factors

*Certain characteristics of patients can increase their chances of experiencing ADRs. These include:*

**Age:** *Older adults often face a higher risk due to physiological changes that affect drug metabolism.*

**Gender:** *Hormonal differences may influence how drugs are processed, leading to varying side effects.*

**Genetics:** *Genetic predispositions can make some individuals more susceptible to specific drug reactions. For instance, certain genetic variants affect how well a person metabolizes medications like clopidogrel.*

**Comorbidities:** *Having multiple health conditions increases complexity in treatment, elevating the risk of interactions between different medications.*

## Drug-Related Factors

*Several aspects related to the drugs themselves also play a role in triggering ADRs:*

**Drug Class:** *Certain classes of medications, like opioids or anticoagulants, have higher incidences of serious side effects.*

**Dosage***: Higher doses frequently correlate with an increased likelihood of adverse reactions. For example, taking more than the recommended dosage of acetaminophen raises toxicity risks.*

**Duration of Use***: Long-term use may lead to cumulative side effects or tolerance issues. Chronic use of corticosteroids can result in complications such as osteoporosis.*

**Drug Interactions***: Combining multiple medications without professional guidance can create dangerous interactions. For example, mixing warfarin with NSAIDs heightens bleeding risks. By recognizing these risk factors, you enhance your ability to discuss potential ADRs with healthcare providers and take proactive steps toward safer medication practices.*

# The primary categories of ADRs:

## Type A (Augmented):

*These predictable reactions occur due to the pharmacological properties of the drug. For example, using anticoagulants can lead to excessive bleeding. They include side effects, secondary effects and toxic effects.*

## Type B (Bizarre):

*These are unpredictable and not dose-dependent. An example is an allergic reaction to penicillin, which may cause hives or anaphylaxis.*

## Type C (Chronic):

*Chronic reactions develop over time due to prolonged medication use. Long-term use of corticosteroids may lead to osteoporosis.*

**Type D (Delayed):** *Delayed reactions appear after a significant period following exposure, like certain cancer treatments causing secondary malignancies years later.*

**Type E (End-of-use):** *These reactions happen when a medication is suddenly discontinued, such as withdrawal symptoms from benzodiazepines.*

# Types of Adverse Drug Reactions

*Adverse drug reactions (ADRs) manifest in various forms, each with distinct characteristics and examples. Understanding these types helps you recognize potential risks associated with medications.*

## Type A: Dose-Dependent Reactions

*Type A reactions occur due to the pharmacological effects of a drug. These reactions are predictable and generally correlate with the dosage. Common examples include:*

**Sedation from antihistamines***: Taking higher doses can lead to increased drowsiness.*

**Hypoglycemia from insulin:** *Excessive insulin intake can cause dangerously low blood sugar levels.*

**Gastrointestinal upset from NSAIDs***: Higher doses may result in stomach irritation or ulcers.*

*These reactions are manageable by adjusting the dose or switching medications.*

## Type B: Idiosyncratic Reactions

*Type B reactions are unpredictable and not related to drug dosage. They often arise in specific individuals due to genetic factors or unique sensitivities. Examples include:*

**Allergic reactions to penicillin:** *Some people develop hives, swelling, or even anaphylaxis after exposure.*

**Stevens-Johnson syndrome from carbamazepine:** *This serious skin reaction can occur without any warning.*

**Hemolytic anemia from sulfonamides***: Certain individuals may experience a drop in red blood cells after taking these antibiotics.*

**Aspirin-induced asthma:** *Some individuals experience worsening asthma symptoms after taking aspirin.*

## Predictable Reactions

■ **Side effects:** *These are unwanted pharmacological effects of a drug, that are seen with therapeutic doses, e.g. atropine used in the treatment of heart block also produces dryness of mouth, blurring of vision, urinary retention, etc., which are the side effects.*

■ **Secondary effects***: The primary action of a drug may result in other effects, e.g. immunosuppression by corticosteroids can lead to development of opportunistic infections, e.g. oral candidiasis.*

■ **Toxic effects:** *These are the effects of a drug, which are either due to overdosage or chronic use, e.g. bleeding due to chronic use/overdosage of anticoagulants and nephrotoxicity with aminoglycosides especially in patients with renal failure.*

# Unpredictable Reactions

■ **Drug allergy:** *It is an abnormal response (local or systemic), mediated by immune system, to a drug/foreign antigen. Different types of hypersensitivity reactions are:*

*Those associated with humoral antibodies: types I, II and III.*

*Those associated with cell-mediated immunity: type IV (delayed hypersensitivity).*

*Type I hypersensitivity (immediate type, anaphylactic) reactions*
*It is a rapidly occurring reaction, hence called immediate hypersensitivity reaction. The manifestations are itching, urticaria, hay fever, asthma or even anaphylactic shock. Itching, rhinitis and urticaria are treated with antihistamines*

# Drug dependence

*WHO defines drug dependence as 'a state, psychic and sometimes also physical, resulting from the interaction between a living organism and a drug characterized by behavioural and other response that always include a compulsion to take the drug on a continuous or periodic basis in order to experience its psychic effects and sometimes to avoid the discomfort of its absence', e.g. opioids, alcohol, barbiturates, amphetamine, etc. The dependence could be psychological or physical.*

*A.* **Psychological dependence***: There is an intense desire to continue taking the drug as the patients feel that their well-being depends on the drug.*

*B.* **Physical dependence***: Repeated drug use produces physiological changes in the body, which makes continuous presence of the drug in the body necessary to maintain normal function. Abrupt stoppage of the drug results in an imbalance wherein the body has to readjust to the absence of the drug resulting in the development of signs and symptoms known as* **withdrawal syndrome***. The withdrawal signs and symptoms are generally opposite to the effects produced by the drug.*

# Withdrawal Reactions

*Withdrawal reactions arise when stopping a medication suddenly after long-term use. Your body might react strongly as it adjusts back to normal functioning without the substance. Some typical withdrawal symptoms are:*

**Benzodiazepine withdrawal:** *This may include anxiety, insomnia, or seizures.*

**Opioid withdrawal:** *Symptoms often involve nausea, muscle aches, and cravings.*

**Antidepressant discontinuation syndrome***: You might experience flu-like symptoms or mood swings.*

*Recognizing these symptoms aids in managing medication changes effectively.*

## Iatrogenic diseases

*It is physician-induced disease ('iatros' is a Greek word, means 'physician') due to drug therapy, e.g. parkinsonism due to metoclopramide; acute gastritis and peptic ulcer due to NSAIDs.*

## Carcinogenicity and mutagenicity

*The ability of a drug to cause cancer is carcinogenicity and the agent is known as carcinogen. The abnormalities of genetic material in a cell produced by a drug is known as mutagenicity, e.g. anticancer drugs and oestrogens.*

## Photosensitivity reactions

*It is a drug-induced cutaneous reaction (photoallergy/phototoxicity) following exposure to ultraviolet radiation. Sulphonamides cause photoallergy on exposure to light; they produce dermatitis due to immune response (cell mediated). Doxycycline and fluoroquinolones can cause phototoxicity – a local reaction (erythema, blisters) occurs on exposure to UV light. Use of sunscreen and avoidance of exposure to sunlight is advised. Calamine lotion and topical steroids are used for treatment*

## Hepatotoxicity

*Some of the hepatotoxic drugs are isoniazid, rifampicin, pyrazinamide, halothane, paracetamol, etc.*

## Nephrotoxicity

*(Vancomycin, aminoglycosides, cisplatin, cyclosporine, amphotericin B, tetracyclines [Fanconi syndrome], indinavir, gold salts, nystatin, etc. are nephrotoxic drugs)*

## Ototoxicity

*It can occur with aminoglycosides, loop diuretics, cisplatin, etc.*

## Ocular toxicity

*Ethambutol, chloroquine, glucocorticoids, etc. can cause ocular toxicity*

# Teratogenicity

*Certain drugs when given during pregnancy may cross the placenta and produce various dangerous effects in the fetus (Table 1.8). This is called teratogenesis.*

*Administration of drugs during early pregnancy (conception to 16 days) could result in abortion; during 2–8 weeks of gestation, it can affect organogenesis and produce structural abnormalities; during second and third trimester, drugs can affect growth and development of the fetus. Hence, drug administration during pregnancy should be restricted.*

*The US FDA (Food and Drug Administration) had placed drugs in various categories (A, B, C, D, X) depending on the risk of the drug to cause birth defects.*

*Category X drugs (e.g. warfarin, methotrexate) was contraindicated for use during pregnancy as risk to fetus was proven and outweighed benefits of its use.*

# References


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